Retail Sales Performance Analysis

Project Information

- **Project Title:** Retail Sales Performance Analysis
- **Project ID:** DA-011
- **Domain:** Retail & Sales Analytics
- **Current Status:** Phase 1 Completed (Data Cleaning & Setup Pushed to GitHub)

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1. Project Overview

In modern retail, stores generate thousands of transaction records every single day. However, raw transaction rows in a CSV file cannot be used directly by store managers to make business decisions.

The goal of this project is to take raw sales data, clean it, analyze it, and turn it into simple, practical business findings. We focus on answering three core questions:

1. **When does the store make the most money?** (Tracking sales across days, months, and quarters)
2. **Which products drive real profit?** (Finding top-selling items versus dead stock)
3. **How do customers shop?** (Understanding basket sizes, visit frequency, and preferred payment methods)

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2. Project Repository & Links

Asset	Platform / Source	Status / Link
GitHub Repository	GitHub	*https://github.com/Sami-Last-Breath-Programming/Retail-Sales-Performance-Analysis*
Google Colab Notebook	Google Cloud	*Pending*

Asset	Platform / Source	Status / Link
Raw Dataset	CSV Archive	*Pending*

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3. Project Team Members

Roll No / ID	Student Name	Assigned Project Role
BCAI2026010	**Abdul Sami**	Data Pipeline & GitHub Management
BCAI2026011	**Nida**	Data Cleaning & Preprocessing
BCAI2026009	**Siya Gupta**	Exploratory Data Analysis
BCAI2026004	**Ayush Singh**	Statistical Modeling & Metrics
BCAI2026006	**Laxmi Joshi**	Data Visualization (Charts)
BCAI2026002	**Arjun Sharma**	Product Category Analysis
BCAI2026007	**Ayush Sharma**	Customer Behavior Analysis
BCAI2026013	**Komal Kumari**	Quality Check & Testing
BCAI2026001	**Paras Gangwar**	Documentation & Report Writing
BCAI2026005	**Manisha Tiwari**	Literature Review & Research
BCAI2026008	**Sharanya Singh**	Dashboard Layout Design
BCAI2026015	**Manish Raj Singh**	Code Review & Repository Sync
BCAI2026003	**Nirbhay Raj Singh**	Verification & Metric Testing
BCAI2026012	**Aditya Priyanshu Verma**	Presentation & Final Review

Methodology & System Architecture

1. Step-by-Step Project Approach

To ensure our findings are reliable and reproducible, our team is following a practical 5-step pipeline:

```
[Raw CSV Data]
       │
       ▼
[Step 1: Data Cleaning] ————> Remove nulls, fix date formats, handle duplicate rows
       │
       ▼
[Step 2: Metric Computation] –> Calculate Gross Revenue, Order Volumes, Average Bill
       │
       ▼
[Step 3: Visual Analytics] ——> Generate time-series plots, category bar charts
       │
       ▼
[Step 4: Business Insights] ——> Suggest inventory changes & promotion timing
```

1. **Data Collection & Ingestion:** We load transaction records stored in CSV format into Python using Pandas dataframes.
2. **Data Cleaning & Validation:** Raw records usually have empty cells, wrong data types, or duplicate bills. We validate every column before calculating anything.
3. **Exploratory Data Analysis (EDA):** We group records by month, category, and payment type to see patterns that are invisible in raw spreadsheets.
4. **Data Visualization:** We convert raw numbers into readable visual charts (bar charts, line graphs, and heatmaps) so non-technical managers can understand them immediately.
5. **Actionable Recommendations:** We conclude with clear suggestions, such as which products to restock before weekend peaks.

2. Technology Stack

Tool / Library	Category	Practical Purpose in this Project
Python	Programming Language	The primary language used to write all analysis scripts and data logic.
Pandas	Data Processing	Loads the CSV dataset, filters rows, joins tables, and computes aggregates.
NumPy	Numerical Library	Performs array transformations and high-speed mathematical operations.
Matplotlib	Visualization	Plots basic graphs including line trends, bar charts, and histograms.
Seaborn	Statistical Visuals	Creates clean heatmaps and correlation charts for customer shopping patterns.
Google Colab / Jupyter	Development Environment	Interactive environment where code, outputs, and explanations live together.
Git & GitHub	Version Control	Keeps our codebase backed up, tracks team commits, and maintains code history.

Current Progress & GitHub Implementation

1. Pushed to GitHub

```
Retail-Sales-Performance-Analysis/
├── readme.md
├── summary.pdf
```